PRASUN JHA

Integrated BS-MS, Mathematics & Computing, Birla Institute of Technology, Mesra
prasunjha080806@gmail.com | +91-8789249770 | github.com/prasunjha8 | Jamshedpur, India

Third-year Mathematics & Computing undergraduate focused on robotics, reinforcement learning, autonomous navigation and control. Builds end-to-end robotic systems spanning perception, decision-making, simulation and embedded hardware.

Education

Birla Institute of Technology, Mesra Aug 2024 – Jul 2029
Integrated BS-MS in Mathematics and Computing
Coursework: Linear Algebra, Probability, Differential Equations, Optimization, Data Structures & Algorithms, OOP

Research & Technical Projects

Vision-Guided Socially Aware Navigation — Ongoing Research [GitHub] 2026 – Present
- Built a camera-only assistive navigation stack in ROS 2 Humble and Gazebo Fortress: an Ackermann-drive robot in a simulated hospital corridor with hazard detection and teleoperation, using no LiDAR or depth sensing.
- Simulated crowds with ORCA (RVO2) pedestrian nodes and added an Open-RMF Airport Terminal crowd stress-test.
- Developing a PPO baseline (Stable-Baselines3) on camera/OpenCV features and testing VLM-derived scene semantics as observations or reward shaping, evaluated on proxemics violations, collisions and time-to-goal.

Fuzzy Robot Coordinator — Multi-Robot Task Allocation [GitHub] 2026
- Built an explainable allocator that scores robot–task suitability from battery, distance, workload, payload ratio and urgency through eight Sugeno-style fuzzy rules, compared against a weighted mathematical baseline.
- Solved exact one-to-one assignment maximizing urgency-weighted suitability under payload and feasibility constraints.
- Integrated the planner with a MuJoCo warehouse of three differential-drive robots with autonomous job execution and battery-aware charging (suspend at 25%, recharge to 85%); added a browser dashboard and 17 automated tests.

RL Self-Balancing Motorcycle — ROS & Gazebo [GitHub] Jun 2025 – Aug 2025
- Trained a Soft Actor-Critic (SAC) agent to self-stabilize a dynamically unstable two-wheeled motorcycle and navigate it to target waypoints in Gazebo 11 with ROS Noetic.
- Built a custom Gym environment bridging Stable-Baselines3 and Gazebo via ROS topics/services, run in Docker.

TortoiseBot Navigation Stack — ROS 2 [GitHub] 2026 – Present
- Architecting a distributed ROS 2 stack for the RigBetel Tortoise robot: a Raspberry Pi 4 edge node (micro-ROS, LiDAR and camera drivers) and a Pi 5 Docker (Jazzy) compute node communicating over DDS.
- Built the Gazebo simulation with URDF/TF2 tree, differential-drive control and teleoperation; configuring SLAM Toolbox and Nav2 costmaps for autonomous navigation.

Flight Simulation Control Systems — JSBSim [GitHub] Apr 2026 – Present
- Built a real-time flight-dynamics pipeline in JSBSim (Cessna 172P and custom RC-plane models) with FlightGear visualization and Python control over UDP.
- Designed and tuned PID controllers for stable flight under disturbances, logging altitude, airspeed and pitch.

Inverted Pendulum on a 4-Wheel Cart — Embedded Control [GitHub] 2024
- Implemented real-time balancing with an ESP8266, MPU6050 IMU, Kalman-filter angle fusion and PID control driving four geared DC motors through an L298N driver.
- Built a Wi-Fi web interface for live PID tuning without reflashing firmware.

Technical Skills

Languages	Python, C++, C
Robotics	ROS 2, Gazebo, MuJoCo, Nav2, SLAM Toolbox, TF2, micro-ROS, CoppeliaSim, ORCA
ML / RL / CV	PyTorch, Stable-Baselines3, PPO, SAC, Q-learning, Gymnasium, OpenCV, YOLOv8, VLMs
Control	PID, Kalman Filtering, Fuzzy Logic (Sugeno), Task Allocation, Optimization
Embedded	ESP32, ESP8266, Arduino, Raspberry Pi, Jetson Orin Nano, Linux, Docker, Git

Achievements

- **Zonal Runner-up**, Ramakrishna Mission Innovation Challenge — wearable assistive technology prototype.
- **Zonal Runner-up**, Dassault Systèmes Innovation Program — foldable 3D-printed helmet design.
- **High Commendation**, Sokovia Accords MUN; **2nd Place**, Pantheon Quiz (UNESQUO); **Special Mention**, Online Model UN (UNESCO committee).

Leadership & Open Source

- **General Secretary, Robolution** (Robotics Club, BIT Mesra) — leading robotics project architecture, technical coordination, sponsorship strategy and junior-member training.
- **GSoC 2026 Proposals, JdeRobot** — submitted proposals for a C++ exercise extension and a deep-learning-based power-tower inspection tool.
- **Events Coordinator, UNESQUO** (Quiz & Debate Society) — organizing competitive debate and quiz events.